

Multi-Node Sovereign Mesh Blueprint & Cloud RLVF Flywheel

Heterogeneous Cluster Orchestration, Asymmetric IAM & Continuous Adaptation

Derived from: [4-ARCHITECTURE-cont.-\(6-files\) \(#2-AESOP XI: Sovereign Edge-Computing Architecture \(Final Master Specification\)., AESOP XI Architecture Blueprint Update, Architecture document \(needs edit\).txt\).](#)

Status: Canonical System Blueprint | Clean Markdown Layer

Target Repository: [data_vault/02_wiki_md/](#)

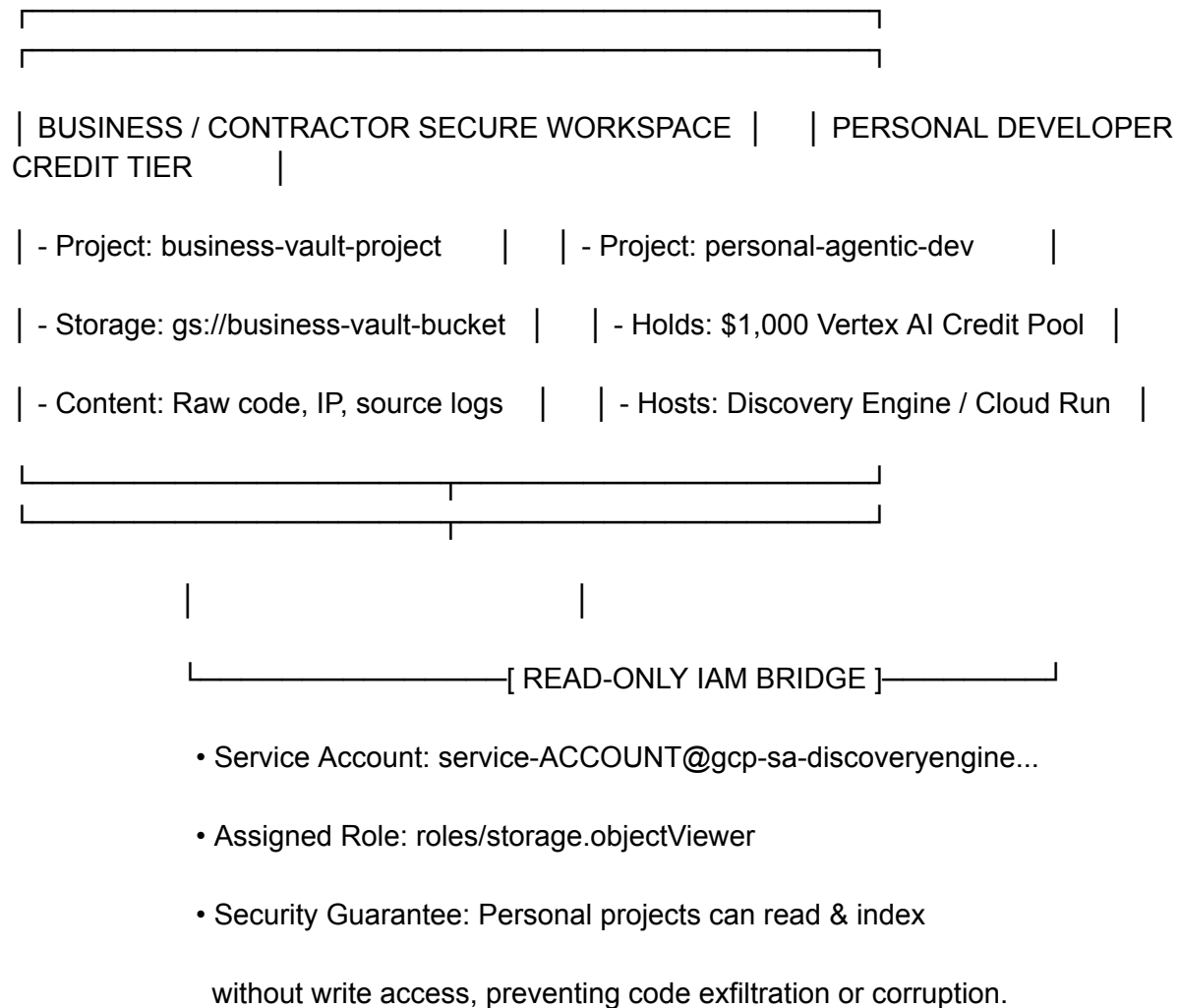
1. The P2P Compute Topology & Mesh Network

The ecosystem links three local physical devices and cloud compute instances over an encrypted Tailscale subnet:

- **Node Alpha (Mobile Engine):** Motorola Razr Ultra 2025. Runs Snapdragon 8 Elite Hexagon NPU (v79), 16GB RAM, Horizons UI + Æsc + Æyre daemons, mem0 episodic cache, local SQLite database, and port 20128 OmniRoute proxy.
 - **Node Beta (Compute Server):** NVIDIA Jetson Orin Nano Super (8GB). Dedicated CUDA tensor cores (60–70 TOPs), 500+GB NVMe SSD. Houses PostgreSQL 16+ OB1 database, Reasoning Bank ledger, Home Assistant Auditor daemon, and the sealed [.incognito_red_sandbox/](#).
 - **Node Gamma (Display Workstation):** Rubik Pi 3 Dragonwing (Qualcomm SoC / 14+ TOPs). Drives dual-monitor hardware display output, keyboard/mouse hub, system telemetry visualizer, and IT help desk manual server.
 - **Node Delta (Enterprise Cloud):** Google Cloud Platform. \$1,000 Vertex AI Agent Builder credits pool for high-compute vector search, Unsloth/GRPO fine-tuning, and long-context verification.
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2. The Asymmetric Cross-Account IAM Handshake

To utilize personal developer credit pools while protecting business data and proprietary code from exposure, the system implements an asymmetric IAM bridge:



3. The Continuous RLVR Learning Flywheel

- Daily Driver Execution:** Node Alpha logs task executions, tool calls, and model outputs into local JSONL buffers.
- End-of-Day Sync:** At end-of-day, the Home Assistant daemon syncs daily traces to Node Beta over Tailscale.
- Red Agent Audit:** The air-gapped Red Auditor evaluates the batch against `nope_data_bank.json` inside `.incognito_red_sandbox/`.
- Cloud Export:** Verified batches (+1.0) are pushed to Google Cloud Storage (`gs://<repo>-rlvr/`).
- Model & Prompt Optimization:** Vertex AI / Compute instances run automated RLVR / GRPO fine-tuning, pushing refined prompt layers back to the edge via Continual Harness.